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YEARS Algorithm for Diagnosis of Suspected Pulmonary Embolism in Patients With Cancer

A Randomized Clinical Trial

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IMPORTANCE Although the YEARS algorithm is a safe and efficient way to rule out acute pulmonary embolism (PE), robust evidence on its accuracy in patients with cancer is lacking, and current guidelines suggest proceeding directly to computed tomographic pulmonary angiography (CTPA).

OBJECTIVE To compare the safety and efficiency of the YEARS algorithm with CTPA only to rule out acute PE in patients with active cancer.

DESIGN, SETTING, AND PARTICIPANTS The Hydra study was an open-label, randomized, investigator-initiated, noninferiority trial with blinded central outcome adjudication conducted from August 22, 2019, to August 21, 2025, the date of final follow-up. Patients with active cancer and suspected acute PE were recruited from emergency departments or medical units in 21 hospitals in the Netherlands, Italy, Switzerland, Belgium, France, and Spain.

INTERVENTIONS Patients were randomly assigned in a 1:1 ratio to receive diagnostic management by the YEARS algorithm (n = 352)—consisting of assessing YEARS items, D-dimer levels, and performing risk-dependent CTPA—or by CTPA only (n = 346).

MAIN OUTCOMES AND MEASURES The primary outcome was centrally adjudicated symptomatic venous thromboembolism or (possible) PE-related death within 90 days after ruling out PE at baseline, assessed in a per-protocol noninferiority analysis with a 2.6% margin for the upper bound of a 1-sided 99.9% CI. The key secondary outcome was the proportion of negative CTPA results at baseline, assessed in a superiority analysis.

RESULTS A total of 698 patients were randomized (median age, 65 years [IQR, 56-72 years], 422 female [60%]), and 104 patients (15%) had PE diagnosed at baseline. One patient was lost to follow-up. Of those in whom PE was considered excluded, 5 of 282 patients (1.8%) in the per-protocol YEARS group vs 15 of 273 patients (5.5%) in the per-protocol CTPA-only group had a primary outcome (absolute risk difference, -3.7%; 99.9% CI, -8.8% to 1.4%; $P = 3.4 \times 10^{-5}$ for noninferiority). In the intention-to-diagnosis analysis, the absolute risk difference between the YEARS algorithm and CTPA only was -2.6% (99.9% CI, -7.5% to 2.4%; $P = 5.9 \times 10^{-4}$ for noninferiority). Diagnostic management of PE was carried out for 77 of 352 patients (22%) in the YEARS group without CTPA. No difference in the proportion of negative CTPA ($P = .93$) was observed between the groups.

CONCLUSIONS AND RELEVANCE In patients with cancer and suspected PE, a diagnostic strategy using the YEARS diagnostic algorithm was as safe as using CTPA only, thus, obviating the need to perform CTPA in 22% of patients.

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Patients with cancer face an increased risk of venous thromboembolism (VTE). The 6-month risk for VTE in patients with newly diagnosed cancer is estimated to be 11 times higher than in the general population, for an absolute risk difference of 1.5%.¹ Furthermore, VTE in patients with cancer is associated with decreased quality of life and higher mortality.²

Computed tomographic pulmonary angiography (CTPA) is considered the diagnostic standard for acute pulmonary embolism (PE).^{3,4} Due to potential harms and costs associated, such as radiation exposure and a prolonged emergency department stay, several clinical decision rules combined with D-dimer testing have been developed to safely exclude PE without CTPA.⁵⁻⁹ The YEARS diagnostic algorithm assesses pretest probability based on 3 clinical items and applies a probability-dependent D-dimer threshold. If the D-dimer level is below the threshold, acute PE can be safely ruled out; if it exceeds the threshold, CTPA is performed.⁸ The YEARS algorithm has been shown to be safe and efficient across several health care settings and patient subgroups.^{8,10} Its use in patients with cancer is debated because post hoc analyses have shown that the incidence of new VTE after excluding acute PE based on the YEARS algorithm was higher in patients with cancer than in patients without cancer.¹⁰⁻¹² Consequently, current guidelines refrain from recommending the use of YEARS or similar clinical decision rules in patients with cancer, and usual practice is often to perform CTPA directly without D-dimer testing.¹³⁻¹⁵ To address this uncertainty, the Hydra study was conducted as a randomized noninferiority trial to evaluate whether the YEARS algorithm is as safe as and more efficient for ruling out PE in patients with active cancer and clinically suspected PE than a diagnostic strategy with CTPA only.

Methods

Trial Design

This study was designed as an open-label randomized, investigator-initiated, noninferiority trial with blinded central outcome adjudication. The trial was performed according to the Good Clinical Practice guidelines and the Declaration of Helsinki.¹⁶ The Consolidated Standards of Reporting Trials (CONSORT) reporting guideline were followed.

Trial Setting

A total of 21 hospitals in 6 countries (the Netherlands, Italy, Switzerland, Belgium, France, and Spain) participated. The trial was funded by the INVENT-VTE network and the participating hospitals without involvement of any commercial entity. The ethics committees of all participating institutions approved the study protocol and all amendments. All patients provided written informed consent before enrollment. A protocol article has been published reflecting the status of the trial as of June 2025,¹⁷ and the study protocol is available in [Supplement 1](#).

Eligibility Criteria

Patients 18 years or older with active cancer (any cancer other than basal cell or squamous cell carcinoma of the skin) and

Key Points

Question Does the YEARS diagnostic algorithm safely and efficiently rule out acute pulmonary embolism (PE) in patients with active cancer?

Findings In this randomized clinical noninferiority trial, the YEARS algorithm's diagnostic failure rate was 1.8% compared with 5.5% with computed tomographic pulmonary angiography (CTPA) only, for an absolute risk difference of -3.7% with the upper boundary of the confidence interval not exceeding the noninferiority margin of 2.6%. Using the YEARS algorithm obviated the need to perform CTPAs in 22% of patients.

Meaning The YEARS algorithm is as safe as a CTPA-only approach to exclude PE in patients with active cancer and avoids CTPA scans.

a clinical suspicion of acute PE were eligible to participate. Active cancer was defined as a cancer diagnosis confirmed by histology or strongly suspected by a clinician within 6 months of study inclusion; ongoing cancer treatment at the time of inclusion; cancer treatment received within 6 months before randomization; or the presence of metastatic disease. Patients were excluded if they required therapeutic anticoagulation for reasons other than suspected PE, had previously participated in the study, underwent diagnostic tests for confirmation of PE found on a prior CT scan, were hemodynamically unstable, had contraindications to CTPA, were enrolled in a concurrent study on thromboprophylaxis, or had a life expectancy of less than 3 months. Patients were recruited from emergency departments, outpatient clinics, and during hospitalization.

Diagnostic Strategies

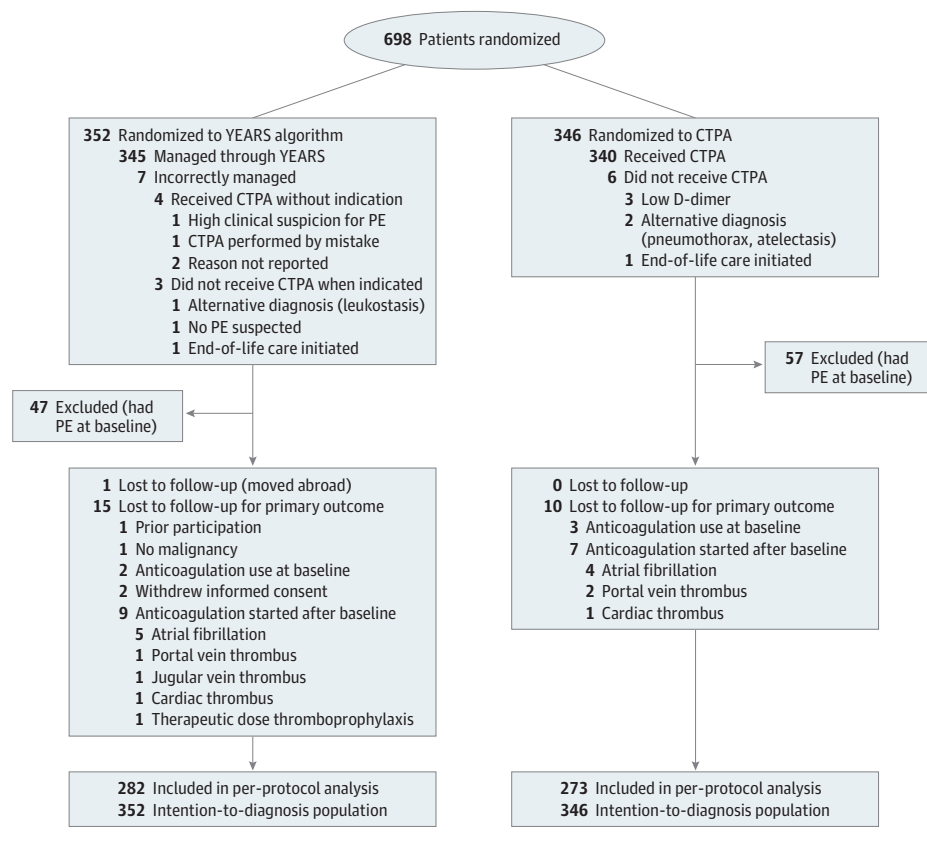
Randomization was conducted centrally with the use of an interactive web-response system with stratification according to participating hospitals. Patients were randomly assigned in a 1:1 ratio to either diagnostic management by the YEARS algorithm or diagnostic management by CTPA only ([Figure 1](#)).

Patients allocated to the YEARS algorithm group were managed by the YEARS algorithm ([Figure 2](#)). If a patient had none of the YEARS items, PE was considered excluded if the D-dimer was lower than 1000 ng/mL. If the patient had 1 or more YEARS items, PE was considered excluded if the D-dimer was lower than 500 ng/mL. In all other circumstances, a CTPA was performed according to standard local clinical practice. The number of YEARS items a patient had was determined by the treating physician before the D-dimer result was known. All patients allocated to the CTPA-only group underwent a CTPA, performed according to standard local clinical practice.

Outcomes

The primary outcome was objectively proven, symptomatic PE, symptomatic upper or lower extremity deep vein thrombosis (DVT) or (possible) PE-related death during the 90-day follow-up period in patients with PE excluded at baseline. Patients were instructed to contact the study physician or the local emergency department in case of signs or symptoms suggestive of VTE during follow-up. Every patient also received

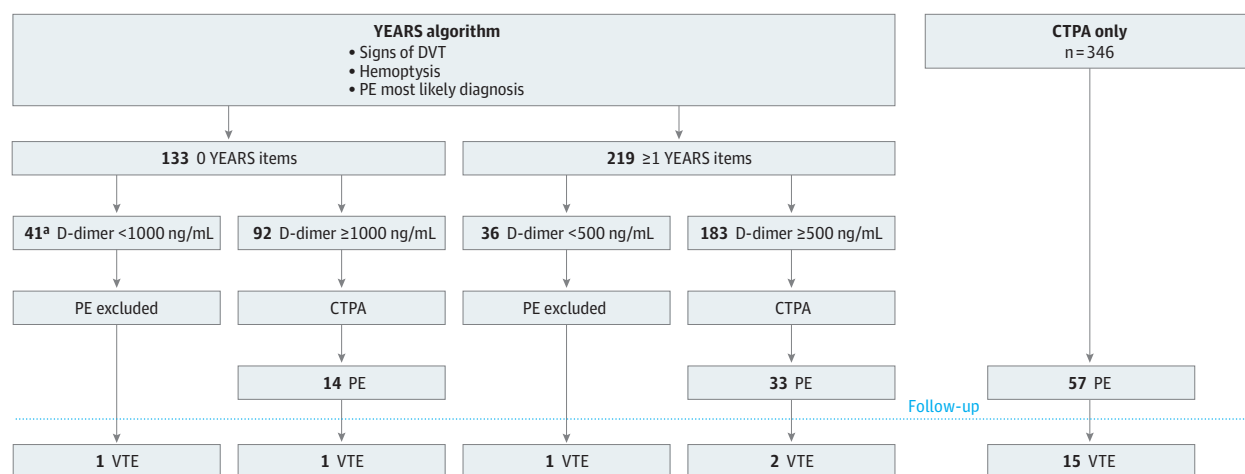
Figure 1. Flow Diagram of Patients in the Hydra Study



The number of patients screened for the study is unknown, mostly because accurate patient screening was not possible during the COVID-19 pandemic. This was not appropriately reinstated after the pandemic in most of the hospitals.

CTPA indicates computed tomographic pulmonary angiography; PE, pulmonary embolism.

Figure 2. Algorithm of the Number of Pulmonary Embolism Diagnoses at Baseline and Venous Thromboembolism Diagnoses During Follow-Up for Both Treatment Groups



^aOf whom 21 patients had a D-dimer higher than 500 ng/mL.

Not included in this figure are 1 incidental VTE in the YEARS group and 3 incidental VTEs in the CTPA-only group.

CTPA indicates computed tomographic pulmonary angiography; DVT, deep vein thrombosis; PE, pulmonary embolism; VTE, venous thromboembolism.

a follow-up telephone call at 90 days (± 7 days) to evaluate the occurrence of signs and symptoms suggestive of VTE during the 3-month follow-up. If the patient could not be reached, the general practitioner and/or primary oncologist was con-

tacted, and the medical records were searched for the possible occurrence of VTE. Incidental VTE was defined as thromboembolism that was detected by means of imaging tests performed for reasons other than clinical suspicion of VTE.

All suspected VTE events during follow-up were adjudicated by an independent committee whose members were blinded to diagnostic allocation, using predefined criteria (Supplement 1). All deaths during follow-up were adjudicated using the International Society on Thrombosis and Haemostasis (ISTH) definition for PE-related death and classification of the cause of death (Supplement 1).¹⁸ Both category A (certainly PE-related) and category B (possibly PE-related) deaths were included in the primary outcome.

The key secondary outcome was efficiency, defined as the proportion of negative CTPAs relative to the number of CTPAs performed.

Statistical Analysis

The trial hypothesis was that the YEARS algorithm was not inferior to management with CTPA with a noninferiority margin of 2.0%. The initial noninferiority margin was based on the original YEARS study,⁸ which found a VTE incidence of 2.6% in the cancer subpopulation during the 3-month follow-up. In this high-risk setting, a margin of 2.0% was considered acceptable. Originally, a sample size of 1566 patients was needed for the per-protocol analysis to have 80% power at a 1-sided type I error of $\alpha = .05$ to show noninferiority of the YEARS algorithm, under the hypothesis that there would be no between-group difference.¹⁷

For the analysis of the primary outcome, we used a per-protocol approach to reduce the risk of type I error.¹⁹ The criteria for inclusion into the per-protocol analysis were exclusion of PE at baseline, confirmed malignancy at baseline, correct application of the allocated diagnostic strategy, no prior participation in the study, complete follow-up, and no therapeutic anticoagulation use during follow-up (Supplement 1). Noninferiority was concluded if the upper bound of the CI of the absolute risk difference at a 1-sided α level of .05 fell below the noninferiority margin of 2.0%.

For the efficiency outcome, we used an intention-to-diagnose approach, which included all patients who were randomized and provided informed consent. Superiority was assessed with a Pearson χ^2 test with an α level set at .05.

Even though a life expectancy of less than 3 months was an exclusion criterion, we observed an overall mortality of approximately 20% during the follow-up. This unexpectedly high mortality was identified because all deaths were reported to the ethics committee and could have been an indicator of a higher rate of PE-related death than originally anticipated. In response, we performed an interim analysis to assess the incidence of VTE, to adjust the noninferiority margin (a higher incidence of VTE during follow-up implicates a higher acceptable noninferiority margin) and sample size. If noninferiority could be demonstrated, recruitment of patients would be stopped. We established a prespecified noninferiority margin for each potential VTE incidence during follow-up before adjudicating the primary end points (see Supplement 1). The noninferiority margin was readjusted separately for both the interim analysis and the final analysis.

After adjudication of the primary end points of all patients included up to July 25, 2024, by the independent blinded central adjudication committee, the noninferiority margin was

adjusted to 2.9% based on an interim incidence of the primary end point of 5.9% in the CTPA-only group. For the interim noninferiority analysis, an adjusted α level of 5.7×10^{-4} was used, based on the proportion of patients enrolled.¹⁹ The interim analysis demonstrated noninferiority of the YEARS algorithm group compared with the CTPA-only group ($P = 1.1 \times 10^{-5}$), after which trial recruitment was stopped. Follow-up for all patients already randomized was completed. For the final analysis of all patients, an adjusted α level of .0013 was used based on the proportion of the originally calculated sample size enrolled.¹⁹

No imputation for missing data was performed. We used R version 4.2.1 (R Foundation for Statistical Computing) for all statistical analyses.

Results

Patient Characteristics

From August 22, 2019, to May 23, 2025, 698 patients were randomized (median age, 65 years [IQR, 56-72 years], 422 female [60%]). A total of 352 patients were allocated to the YEARS group and 346 to the CTPA-only group (Figure 1). Baseline characteristics were comparable (Table). A total of 104 patients (15%) had PE diagnosed at baseline; 47 (13%) in the YEARS group and 57 (16%) in the CTPA-only group. In the YEARS group, 14 (4.0%) were diagnosed with PE up to the central pulmonary artery, 21 (6.0%) up to the segmental branches, and 12 (3.4%) only in subsegmental arteries. In the CTPA-only group, this was 17 (5.0%), 29 (8.4%), and 11 (3.2%), respectively. One patient in the YEARS group was lost to follow-up after 16 days because of moving abroad, no signs of VTE had been present during follow-up. No patients in the CTPA-only group were lost to follow-up (Figure 1). Of the patients randomized to management by the YEARS algorithm, 58% had the item "PE most likely diagnosis" scored, and 46% as the only item.

Primary Outcome

Of the 555 patients included in the per-protocol analysis, 282 were in the YEARS group and 273 in the CTPA-only group. The reasons for exclusion from the per-protocol analysis are shown in Figure 1.

In the YEARS group, 5 patients (1.8%) were diagnosed with VTE during the 3-month follow-up: 3 patients had a nonfatal PE and 2 patients had a possible PE-related death. In the CTPA-only group, 15 patients (5.5%) were diagnosed with VTE during the 3-month follow-up: 5 patients had a nonfatal PE, 2 patients had a fatal PE, 5 patients had a possible PE-related death, and 3 patients had a DVT (eTable 1 in Supplement 2). According to the predefined criteria in the protocol (Supplement 1), the noninferiority margin was readjusted to 2.6% based on the incidence of the primary end point of 5.5% in the CTPA-only group. The absolute risk difference between the YEARS algorithm and CTPA only was -3.7% (99.9% CI, -8.8% to 1.4%; $P = 3.4 \times 10^{-5}$ for noninferiority). The noninferiority criterion for the YEARS algorithm compared with CTPA only was therefore met.

Table. Demographic and Clinical Characteristics of Patients at Baseline

	No. (%) of patients	
	YEARS (n = 352)	CTPA only (n = 346)
Age, median (IQR), y	66.0 (56.8-73.0)	64.0 (56.0-72.0)
Female	215 (61.1)	207 (59.8)
Male	137 (38.9)	139 (40.2)
Body mass index, median (IQR) ^a	25.3 (22.1-28.9)	25.4 (22.7-29.1)
Smoking history		
No	156 (44.4)	152 (44.2)
Former	156 (44.4)	143 (41.6)
Current	39 (11.1)	49 (14.2)
Antiplatelet therapy	75 (21.3)	60 (17.3)
History of venous thromboembolism	20 (5.7)	17 (4.9)
Recent surgery ^b	43 (12.2)	41 (11.8)
Setting		
Emergency department	269 (76.4)	265 (76.6)
During hospitalization	54 (15.3)	51 (14.7)
Outpatient clinic	29 (8.2)	30 (8.7)
D-dimer assay ^c		
Innovance	139 (39.5)	NA
STA Liatest	130 (36.9)	NA
TinaQuant	54 (15.3)	NA
VIDAS	24 (6.8)	NA
HemosIL	5 (1.4)	NA
Primary cancer site ^d		
Lung	94 (26.7)	94 (27.2)
Genitourinary tract	75 (21.3)	88 (25.4)
Breast	52 (14.8)	47 (13.6)
Hematological	41 (11.6)	43 (12.4)
Upper gastrointestinal tract	36 (10.2)	27 (7.8)
Lower gastrointestinal tract	34 (9.7)	19 (5.5)
Other	14 (4.0)	22 (6.4)
Melanoma skin cancer	6 (1.7)	8 (2.3)
Endocrine	5 (1.4)	4 (1.2)
Brain	3 (0.9)	5 (1.4)
Anticancer therapy in previous 6 mo ^d		
Chemotherapy	193 (54.8)	176 (50.9)
Immunotherapy	91 (25.9)	117 (33.8)
Radiation therapy	74 (21.0)	60 (17.3)
Surgery	62 (17.6)	62 (17.9)
No anticancer therapy	48 (13.6)	44 (12.7)
Hormone therapy	29 (8.2)	24 (6.9)
Cancer stage		
Localized	77 (21.9)	56 (16.2)
Regional	77 (21.9)	79 (22.8)
Distant metastases	169 (48.0)	177 (51.2)
Comorbidities ^d		
Hypertension	107 (30.4)	127 (36.7)
COPD	64 (18.2)	57 (16.5)
Diabetes	44 (12.5)	48 (13.9)
Ischemic heart disease	34 (9.7)	27 (7.8)
Stroke	27 (7.7)	20 (5.8)
Rheumatic or other autoimmune disease	21 (6.0)	16 (4.6)
Heart failure	11 (3.1)	9 (2.6)

(continued)

Table. Demographic and Clinical Characteristics of Patients at Baseline (continued)

	No. (%) of patients	
	YEARS (n = 352)	CTPA only (n = 346)
Presenting symptoms ^d		
Dyspnea	275 (78.8)	282 (81.5)
Chest pain	189 (54.0)	192 (55.7)
Coughing	118 (33.8)	122 (35.7)
Symptoms of the limb(s)	52 (14.8)	36 (10.5)
Syncope	40 (11.9)	34 (10.3)
Palpitations	38 (11.2)	30 (9.1)
Hemoptysis	20 (5.7)	20 (5.8)

Abbreviations: COPD, chronic obstructive pulmonary disease; CTPA, computed tomography pulmonary angiography; NA, not applicable.

^a Body mass index is calculated as weight in kilograms divided by height in meters squared.

^b Surgery in the past 4 weeks.

^c D-dimer assays and manufacturers: Innovance, Siemens Healthineers; STA Liatest, Stago; VIDAS D-Dimer Exclusion II, Biomérieux; HemosIL, Werfen.

^d Not mutually exclusive.

In a scenario, in which the patient lost to follow-up in the YEARS group was considered to have met the primary outcome, the 3-month incidence would have been 2.1% in the YEARS group (6 of 282 patients), with an absolute risk difference between the YEARS algorithm and CTPA only of -3.4% (99.9% CI, -8.6% to 1.8%; $P = 1.2 \times 10^{-4}$ for noninferiority).

In the intention-to-diagnosis analysis, the absolute risk difference between the YEARS algorithm and CTPA only was -2.6% (99.9% CI, -7.5% to 2.4%; $P = 5.9 \times 10^{-4}$ for noninferiority; eTable 2 in Supplement 2), still meeting the noninferiority criterion.

Efficiency Outcome and Other Secondary Outcomes

Of the 352 patients allocated to the YEARS group, 77 patients (22%) were managed without CTPA (Figure 2). The number of negative CTPA results as a proportion of the total number of CTPA performed at the initial testing of patients randomized to the YEARS algorithm was not different from that in patients randomized to the CTPA-only approach (83% vs 83%, $P = .93$ for superiority).

During the 3-month follow-up, 1 incidental PE in the YEARS group and 1 incidental PE and 2 incidental DVTs in the CTPA-only group (eTables 1 and 2 in Supplement 2) were observed. All incidental VTEs were treated with therapeutic anticoagulation. The 3-month all-cause mortality was 21% ($n = 73$) in the YEARS group and 21% ($n = 74$) in the CTPA-only group.

Discussion

This study found that diagnostic management with the YEARS algorithm to exclude acute PE in patients with cancer is as safe as a diagnostic strategy with CTPA only. Importantly, PE could be excluded without CTPA in a relevant proportion of 22% of patients managed according to the YEARS algorithm, saving costs and decreasing exposure to ionizing radiation, the latter of which is of relative importance in this population, and is saving patients from receiving potentially harmful contrast material.^{9,20}

The interim analysis-adjusted upper bound of the CI of the absolute difference between the 2 strategies of 1.4% was well below the adapted noninferiority margin of 2.6%. Moreover, it also remained below the initially chosen noninferiority margin of 2.0%, further supporting the safety of the YEARS algorithm in this setting.

This study found a numerically lower number of VTE in the YEARS group compared with the CTPA-only group. Because the YEARS algorithm cannot, by definition, have superior safety compared with management by CTPA only, we ascribe this observation to statistical variation, in the setting of similar performance of the 2 diagnostic strategies. However, 3, albeit unlikely, potential explanations could be given. First, patients randomized to the YEARS group and managed without CTPA at baseline might have undergone CT scans of the chest for other reasons (eg, for a cancer treatment response evaluation) during follow-up more often than patients randomized to the CTPA-only group. This could have increased certainty in ruling out PE as the primary cause of death, depending on the timing of the CT scan. However, with comparable overall mortality in both study groups, only a numerically slightly higher 4% of patients in the CTPA-only group possibly died of PE. Second, incidental VTEs detected on CT scans, which were not part of the primary outcome, were all treated with anticoagulation, thus reducing the risk of evolving to symptomatic PE. However, 3 of the 4 incidental VTEs identified during follow-up occurred in the CTPA-only group. Third, because the YEARS algorithm includes an explicit assessment for DVT, treating physicians might be less likely to miss DVT at baseline in the YEARS group, leading to more DVT diagnoses in the CTPA-only group during follow-up. However, baseline symptoms suggestive for DVT were recorded and found to be rare in both groups. Together, these 3 explanations are unlikely to account for the observed numerical difference in the primary outcome.

Some other observations deserve comment. First, the 22% of patients whose care was managed without CTPA in the YEARS group was lower than the 48% reported in the original YEARS study, which included mostly patients without cancer,⁸

but similar to the 21% (95% CI, 17% to 25%) reported for patients with cancer and suspected PE in a large, individual-patient data meta-analysis.²¹ Second, the proportion of negative CTPAs at baseline was comparable in both groups. Third, there was no difference between groups in the number of subsegmental PEs diagnosed at baseline, which is contrary to the findings from the original YEARS cohort (with and without cancer). However, the numbers were low and this finding should be regarded as hypothesis generating; in a cancer population in which many patients proceed to CTPA, the absence of this effect is mechanistically consistent.²² Fourth, of all patients in previous studies in which the YEARS algorithm was evaluated in a general population with suspected PE, including those with cancer, 50% to 60% of patients had at least 1 YEARS item, of whom 70% to 80% have the criterion “PE most likely” as a single YEARS criterion.^{8,10} The distribution of the YEARS items therefore seems very comparable between the original YEARS study and the Hydra study.

What do these results mean for clinical practice? First, even though the majority of patients with cancer and suspected PE still have to undergo CTPA, the use of the YEARS algorithm in this setting will result in a substantial annual reduction in the total number of CTPAs performed. Historically, an estimated 20% to 30% of all first VTE events are cancer associated, and this incidence has been increasing over the years.^{1,23} Under the presumption that the D-dimer test is not useful and unnecessarily delays the diagnostic process, clinicians were inclined to immediately perform CTPAs in patients with cancer. Implementing the YEARS algorithm for PE diagnosis in these patients not only will reduce health care costs and the growing workload for radiology departments but will also result in patients spending less time in the hospital, undergoing less radiation, and experiencing less exposure to contrast material. Extrapolating these results, given that the age-adjusted incidence of cancer-associated VTE is 70 per 100 000 adults in the general population, would translate to an estimated annual number of 179 000 cases in the US.²⁴ Because one-third of these cases is PE and the prevalence of suspected PE is 4 to 5 times higher than the number of confirmed cases, the impact on societal level is definitely relevant: extrapolating these data yield an estimated reduction of between 70 000 and 80 000 CT scans annually if the YEARS algorithm were implemented.

Second, although it is perceived that the YEARS item “PE most likely” diagnosis is subjective, it has been consistently reported to be the strongest predictor of PE and was retained in more recent studies in which the Wells score was updated. Assessing the item by both experienced and more junior health care professionals has been shown to be accurate and acceptably reproducible, both in the setting of suspected DVT and PE.²⁵⁻²⁸ Third, the study was not powered to detect differ-

ences in the diagnostic yield between both study groups. We nevertheless expect that on a population level, the yield of CTPAs when the YEARS diagnostic algorithm is applied would indeed be better compared with the CTPA-only diagnostic approach. Fourth, our explanation for mortality being higher than expected is that (1) cancer-associated PE is a strong predictor of mortality and (2) estimating life expectancy is notoriously difficult. Indeed, 21% of this population died, including those diagnosed with PE at baseline. As a result, we included patients with poor life expectancy, and our data are generalizable to the broad cancer population except for those already in palliative care.

To our knowledge, this study is the first randomized trial to compare a validated diagnostic algorithm to rule out acute PE with management using CTPA only specifically in patients with active cancer. Additional strengths of this study include the use of a diagnostic-standard control group, blinded outcome adjudication, near complete follow-up, and inclusion of a substantial number of outpatients and inpatients, increasing the external validity of our findings. Finally, a relevant proportion of patients presented with metastatic cancer, had undergone recent surgery, or were receiving chemotherapy, demonstrating that these findings are applicable for patients with active cancer at high risk of thrombosis.

Limitations

Some limitations also need to be mentioned. First, because we excluded patients with a life expectancy of less than 3 months, these results may be less applicable for these patients. Second, because patients and treating physicians were not blinded to the allocation of the diagnostic strategy, bias could have been introduced. We accounted for this by independent blinded adjudication of all outcome events. Third, the interim analysis could have potentially impacted the reliability of the study results. To avoid this, we performed the interim analysis and also prespecified the noninferiority margins before assessing the final primary outcome, ensuring no influence from interim data. We also applied an adjusted α to account for interim statistical testing. Fourth, we were not able to perform any subgroup analyses for different primary cancer sites and stages due to sample size constraints.

Conclusions

The results of this noninferiority trial demonstrated that the YEARS diagnostic algorithm is as safe as CTPA only in excluding PE in patients with cancer with a low risk of VTE during a 3-month follow-up. Using the YEARS algorithm led to obviating the need for a CTPA in 22% of patients.

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